PART-7

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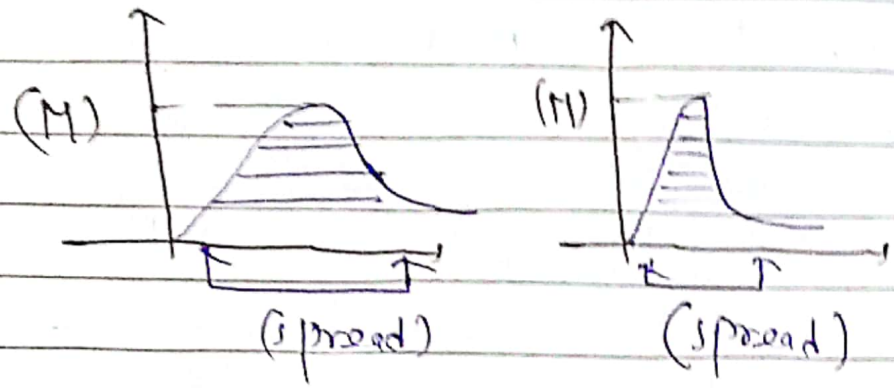
(#) :- Measures of dispersion :-

→ Dispersion is degree of variation of data
→ (spread, variability)

(#) Graph :-



→ Range
→ Inter Quartile Range (IQR)
→ quartile
→ percentile
→ varience
→ std.

(1) Range :-

(Range = Max − Min)

→ sensitive to extreme values
→ it is not consider the distribution of data

→ it is suitable for small dataset

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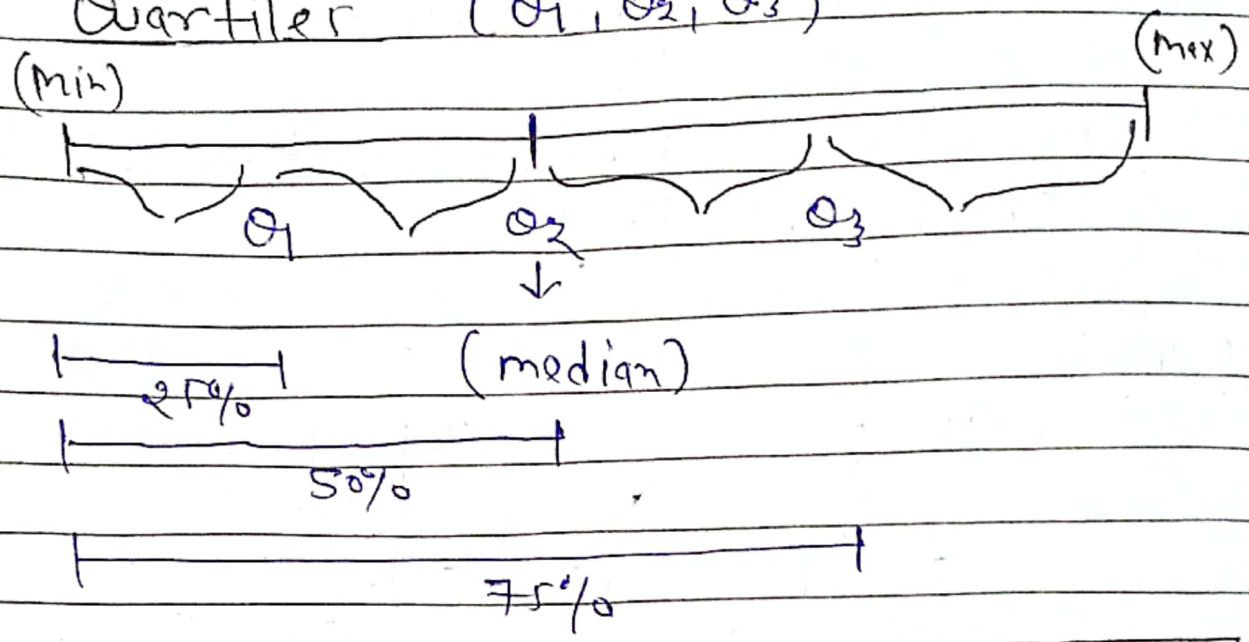
(ii) Quartiles (Q_1, Q_2, Q_3)



(#) $\boxed{Q_1 = \frac{n+1}{4}}$ $\boxed{Q_2 = \frac{n+1}{2}}$ $\boxed{Q_3 = \frac{3(n+1)}{4}}$

(#) Percentile:

$$\boxed{P = \frac{p}{100} \times (n+1)}$$

P: Given percentage
n: no. of observation

(#) IQR:

→ ($IQR = Q_3 - Q_1 = 50\%$ of the data)

→ less sensitive to extreme values

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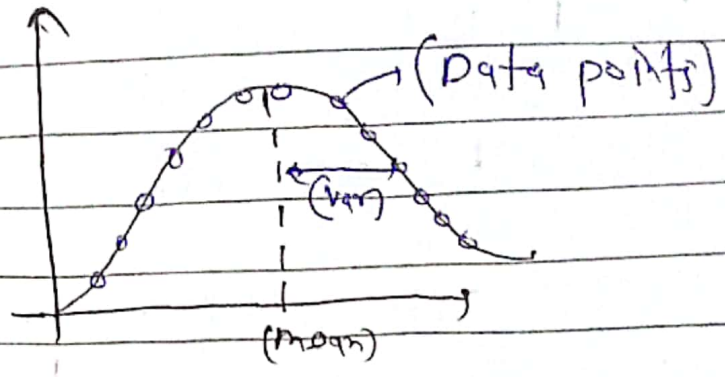
(#) Varience :-

$$V = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$



where:

X_i : Individual data points

μ : mean

(#) Standard deviation (std) :-

$$std = \sqrt{varience}$$

Note :-

std has same calculated UNIT with original dataset

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